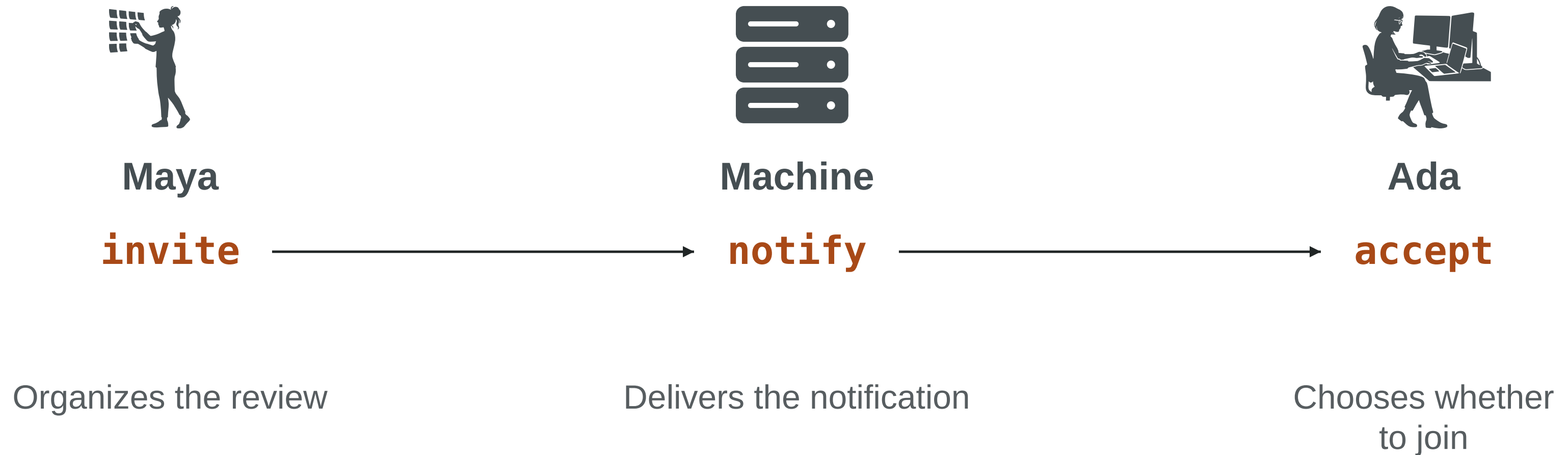


Designing coordination

How people, machines, and LLM agents work together

Inviting Ada to the review

Maya is organizing Thursday's refunds review.
She wants Ada, the payments lead, to join.



PART 1

State and actions

Inviting [Engagement, Guest]

Purpose Ensure mutual consent to participate in an engagement. This **prevents** someone being counted in without agreeing, or turning up uninvited.

Principle Maya invites Ada to the review. Ada accepts and is counted in.

State

a set of Invitations with
 an Engagement
 an invited Guest
 an expiry Date
 a status of
 PENDING, ACCEPTED,
 DECLINED, EXPIRED

Actions

invite(engagement, guest, expiry)
 return (invitation)

accept(invitation)

decline(invitation)

expire(invitation)

Maya requests an invitation



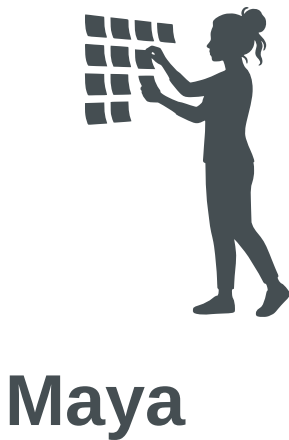
Maya

invite(...)



Inviting

An invitation becomes a record



invite(...) →

Inviting

 Invitation

engagement	the review
invited	Ada
expiry	Wednesday
status	PENDING

The record remains after Maya’s action finishes.

Ada requests acceptance



Maya

Earlier

`invite(...)`



Ada

`accept(invitation)`



Inviting

 Invitation

engagement	the review
invited	Ada
expiry	Wednesday
status	PENDING

Ada changes the invitation



Maya

Earlier

`invite(...)`



Ada

`accept(invitation)`



Inviting

 Invitation

engagement	the review
invited	Ada
expiry	Wednesday
status	ACCEPTED

Ada accepts the invitation Maya created.

1 · What if Ada does not reply?



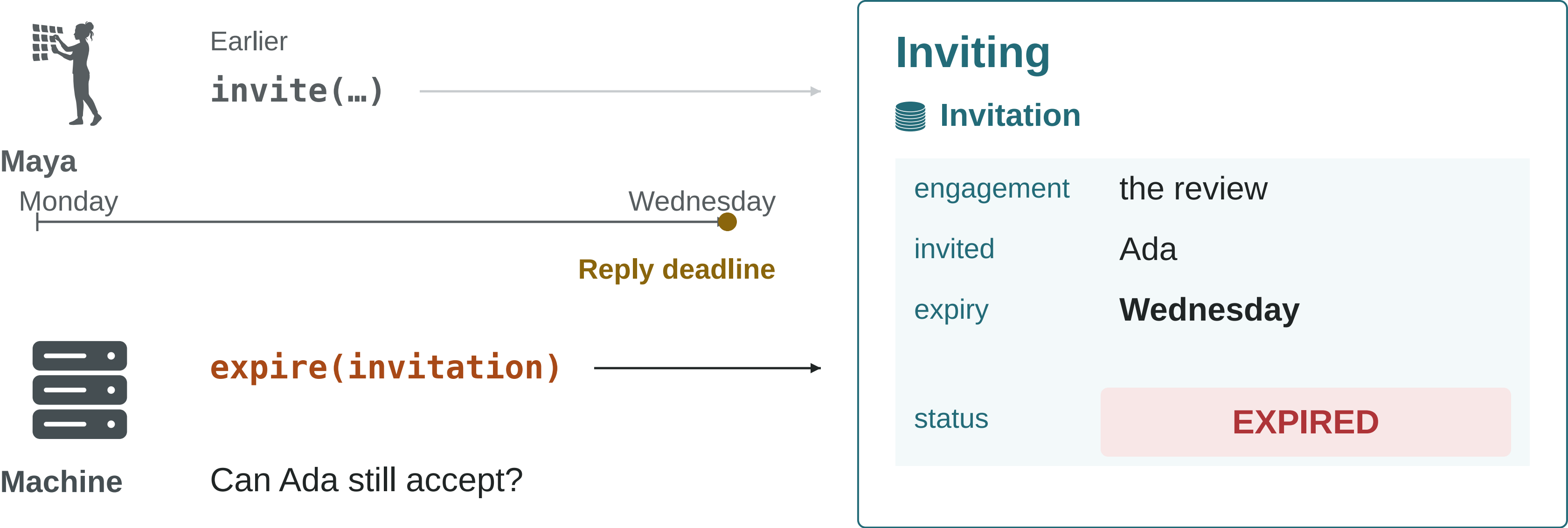
Who calls expire?

Inviting

 **Invitation**

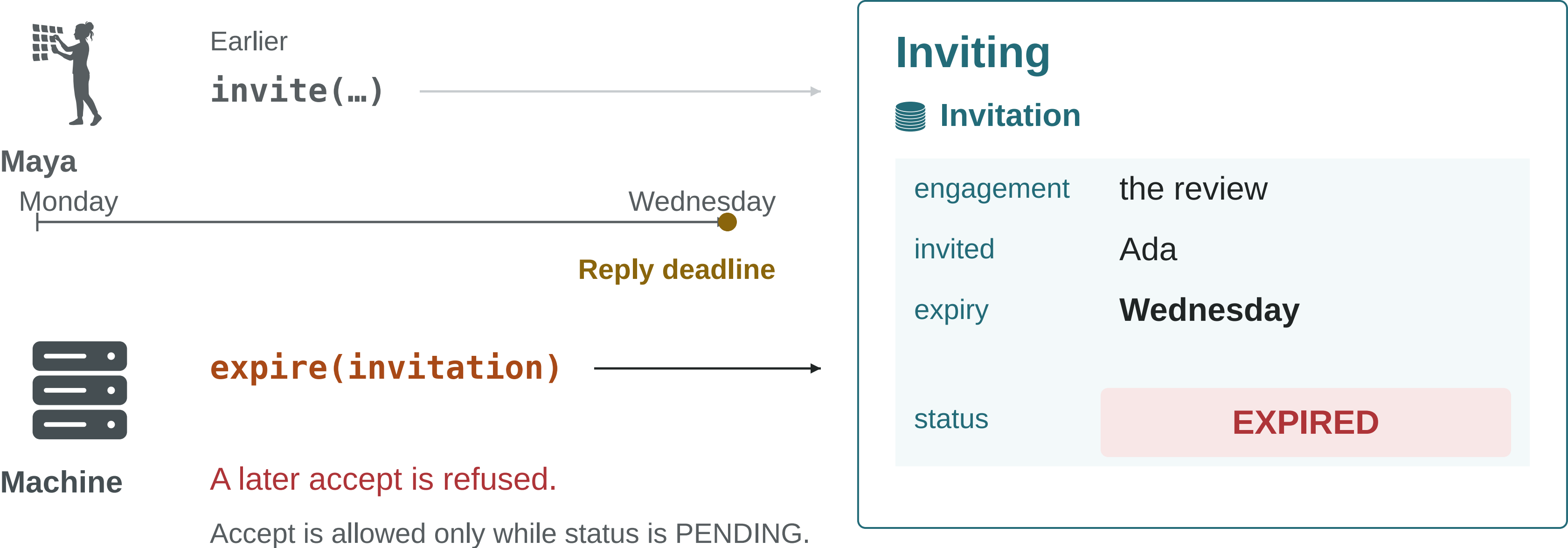
engagement	the review
invited	Ada
expiry	Wednesday
status	PENDING

1 · What if Ada does not reply?



The machine calls expire. Inviting changes the record.

1 · What if Ada does not reply?



The machine calls expire. Inviting changes the record.

2 · Actions carry shared meaning



“I invited **Ada** to **the review**,
with a reply due by **Wednesday**.”

engagement

guest

expiry

the review

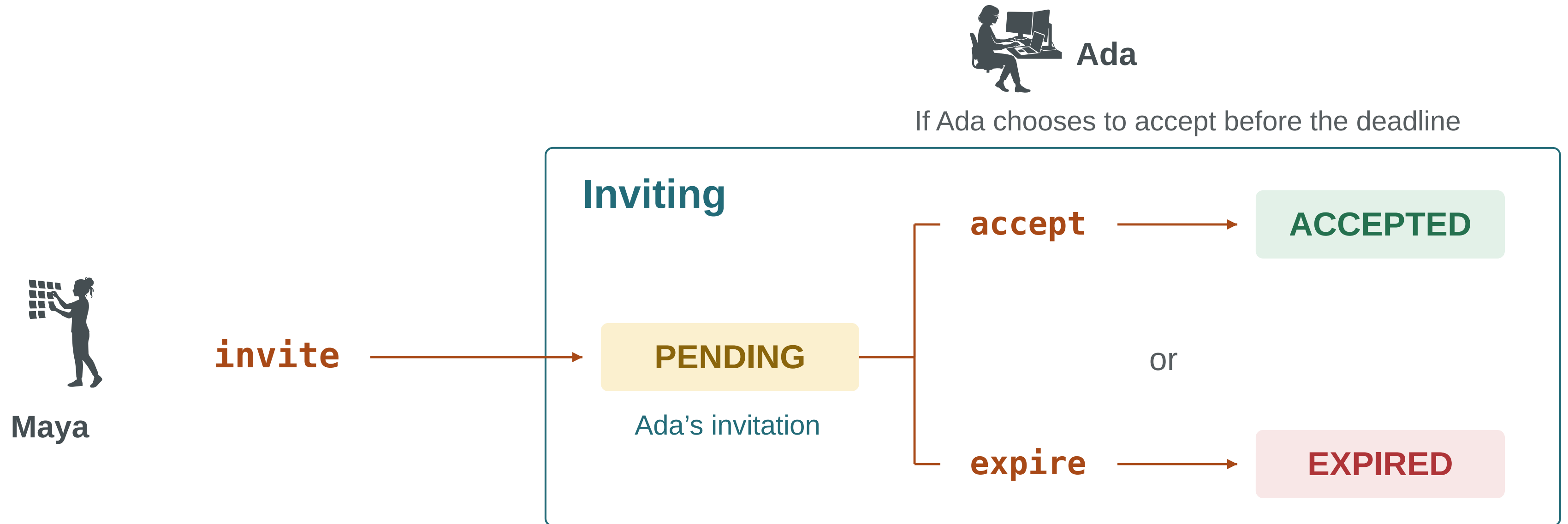
Ada

Wednesday

Inviting.invite(**the review**, **Ada**, **Wednesday**)

What does this action tell the machine?

People and software act through the same concept



 **Ada**
 If Ada chooses to accept before the deadline

If Wednesday arrives without a reply

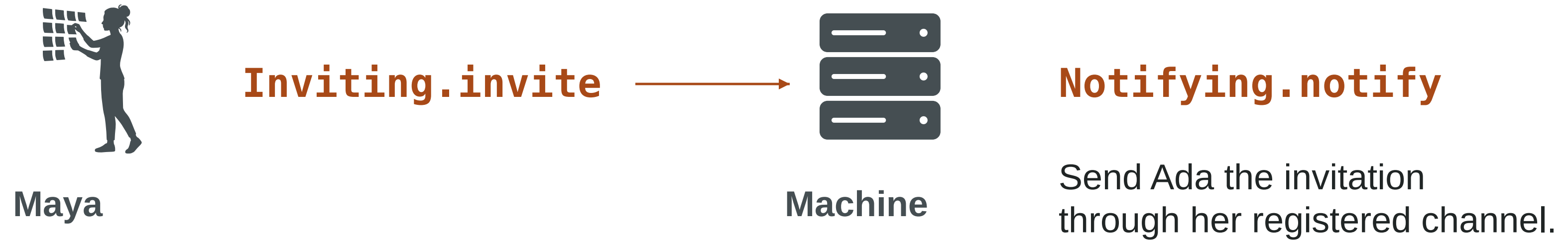
Inviting keeps the request and the guest's answer together.

 **Machine**

PART 2

Reactions and reads

How does an invitation lead to a message?



Notifying [Person]

Purpose Get a message to a person through the channel they registered. This **prevents** a message going to a channel the person never checks, or nobody knowing how to reach them.

Principle Ada registers her phone. A notification for Ada goes to her phone.

State

a set of Persons with
a channel String

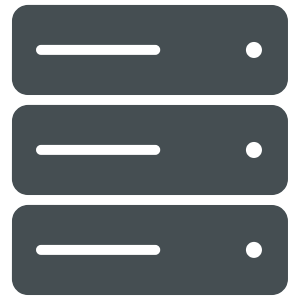
a set of Notifications with
a person Person
a message String

Actions

register(person, channel)

notify(person, message)
return (notification)

A machine reaction



when **Inviting.invite**

where **Check Ada has a registered channel**
Read without changing state.

then **Notifying.notify(Ada, message)**

Ada's reaction includes her judgment



Ada

when **Notifying.notify**
Ada receives the invitation.

where **Inviting**

engagement	the review
expiry	Wednesday

“Can I make this review?”

then **accept(invitation)** or **decline(invitation)**

Ada's reaction includes her judgment



Ada

when

Notifying.notify

Ada receives the invitation.

where

 **Inviting**

engagement	the review
expiry	Wednesday



then

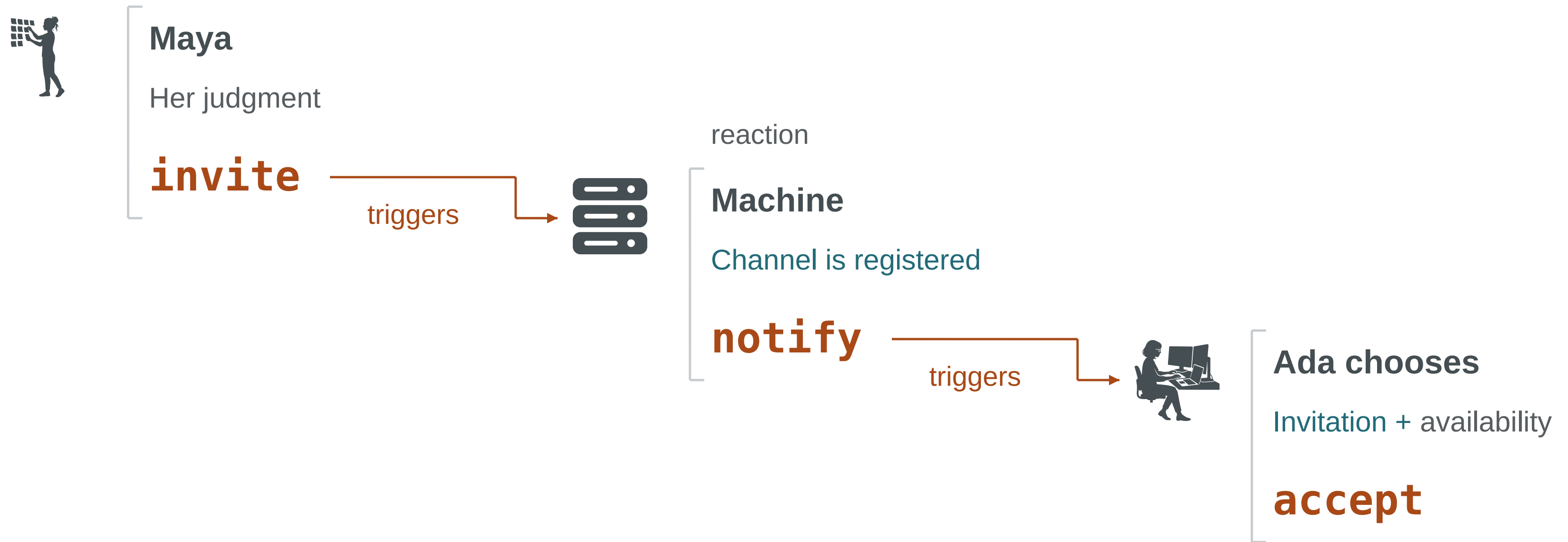
accept(invitation)

or

decline(invitation)

✓ Ada chooses to accept.

3 · Reactions connect through actions

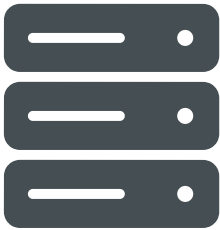


The where clause reads concept state

Previously registered

 **Notifying**

person	Ada
channel	phone



Machine

where

Ada has a
registered channel

The reaction reads the record without changing it.

4 · People read state through a view

 Inviting

engagement	the review
invited	Ada
expiry	Wednesday

The review

Please reply
by Wednesday.

⋮

where Invitation + her availability

 **Ada**

“Can I make
this review?”

Reactions connect actions and read state



PART 3

Delegating work

Man-Computer Symbiosis

‡

IRE TRANSACTIONS ON HUMAN FACTORS IN ELECTRONICS

March

J. C. R. Licklider
1960

Man-Computer Symbiosis*

J. C. R. LICKLIDER†



People

“Men will set the goals, formulate the hypotheses, determine the criteria, and perform the evaluations.”



Computing machines

“Computing machines will do the routinizable work that must be done to prepare the way for insights and decisions in technical and scientific thinking”.

Preparing to think

85%

of his “thinking” time
went into preparation.

His informal self-study, 1957. Published in 1960.

One task: compare speech-intelligibility experiments

Hours

to make differently measured
results comparable



then

Seconds

to draw the needed
conclusion

Maya organizes a refunds review



Maya

Thursday's review

“We need to discuss
returning customers' money.”

Maya identifies the concern:

refunds

Who is responsible for this concern?

Convening [Concern]

Purpose	Set up an occasion and establish what it is about. This prevents people arriving without knowing its purpose, or arriving for an occasion that was called off.
Principle	Maya convenes Thursday’s review and marks refunds as a concern.

State

a set of Occasions with

- a title String
- a time Date
- a description String
- a set of Concerns

Actions

convene(title, time, description)
 return (occasion)

cover(occasion, concern)

drop(occasion, concern)

cancel(occasion)

Where do we find who is responsible?

 Convening

occasion	the review
concern	refunds

Which person should receive an invitation?

We need a record connecting responsibilities to people.

The team’s responsibilities

Payments lead
refunds



Ada

Operations lead
outages



Sam

Rostering [Concern]

Purpose	Record who holds each position and what it is responsible for. This prevents responsibilities being unclear, or two people each assuming the other is responsible.
Principle	Maya appoints Ada as payments lead, responsible for refunds. Sam can look her up.

State

a set of Positions with
 a title String
 a holder Person, or none
 a set of Concerns

Actions

create(title)
 return (position)

assign(position, concern)

appoint(position, person)

relieve(position)

A concern and a subject can be the same thing

Convening [Concern]

Rostering [Subject]



refunds

Your application might call this a Topic.

The shared concern leads to Ada

Maya marks the review with cover(the review, refunds).

 Convening

occasion	the review
time	Thursday
concern	<u>refunds</u>

 Rostering

position	payments lead
holder	Ada
concern	<u>refunds</u>

same
concern

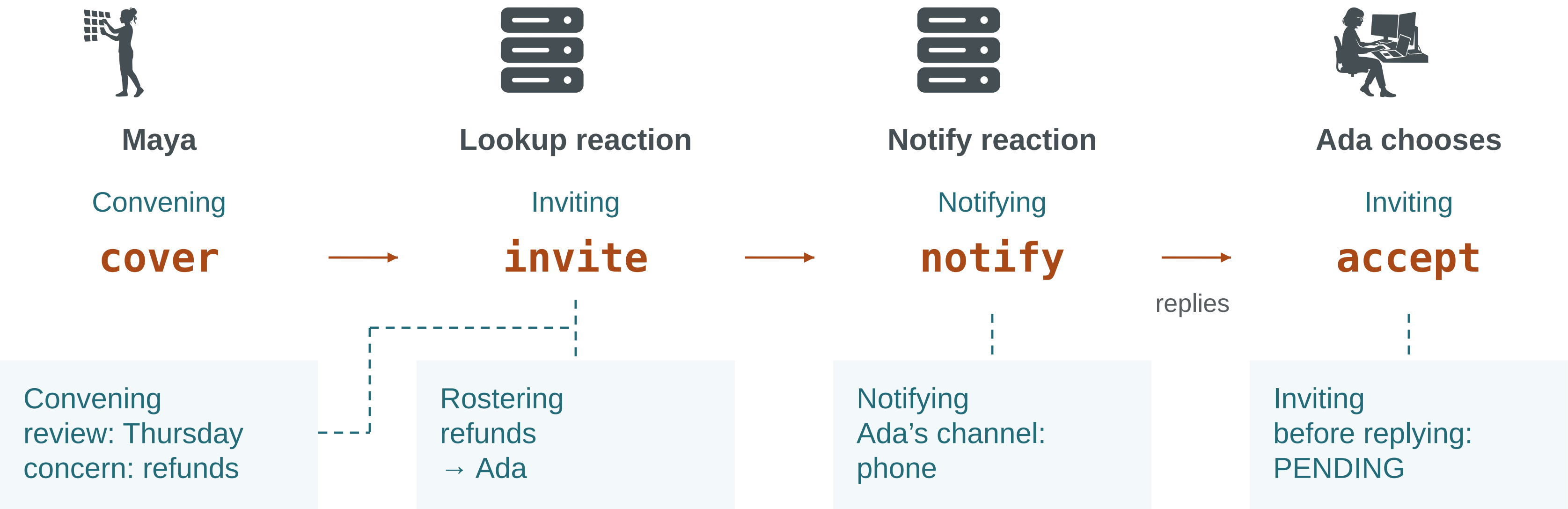


Inviting.invite(the review, Ada, Wednesday)

The machine reads the match and sets the reply deadline.

The review reaches the responsible person

Earlier: Maya records the review and roster. Ada registers her phone.



Maya names the concern. Software handles the chain. Ada decides.

Where might this solution fall short?

The invitation reached Ada. The reactions worked.



Would this always bring
the right people into the room?

Austin contributed, but holds no roster position



Ada

Payments lead

INVITATION RECEIVED



Austin

Built the refunds fix

No position links
Austin to refunds.

Formal responsibility and current contribution are different facts.

Keeping contributions visible takes work

Sam sees an update

Austin: “My refunds patch is merged. Ready for release.”



Sam understands who built the fix.

Who records the update?

Later, Maya plans the review



“Who contributed to refunds?”

No shared record of Austin’s contribution

Each update needs interpretation and upkeep before someone else can look it up.

Repeating the roster lookup

 Rostering

position	payments lead
holder	Ada
concern	refunds



For refunds

Find matching positions.
Invite their holders.

Would running the same
lookup again find Austin?

The lookup only sees recorded responsibility

 Rostering

position	payments lead
holder	Ada
concern	refunds



For refunds

Find matching positions.
Invite their holders.

ADA

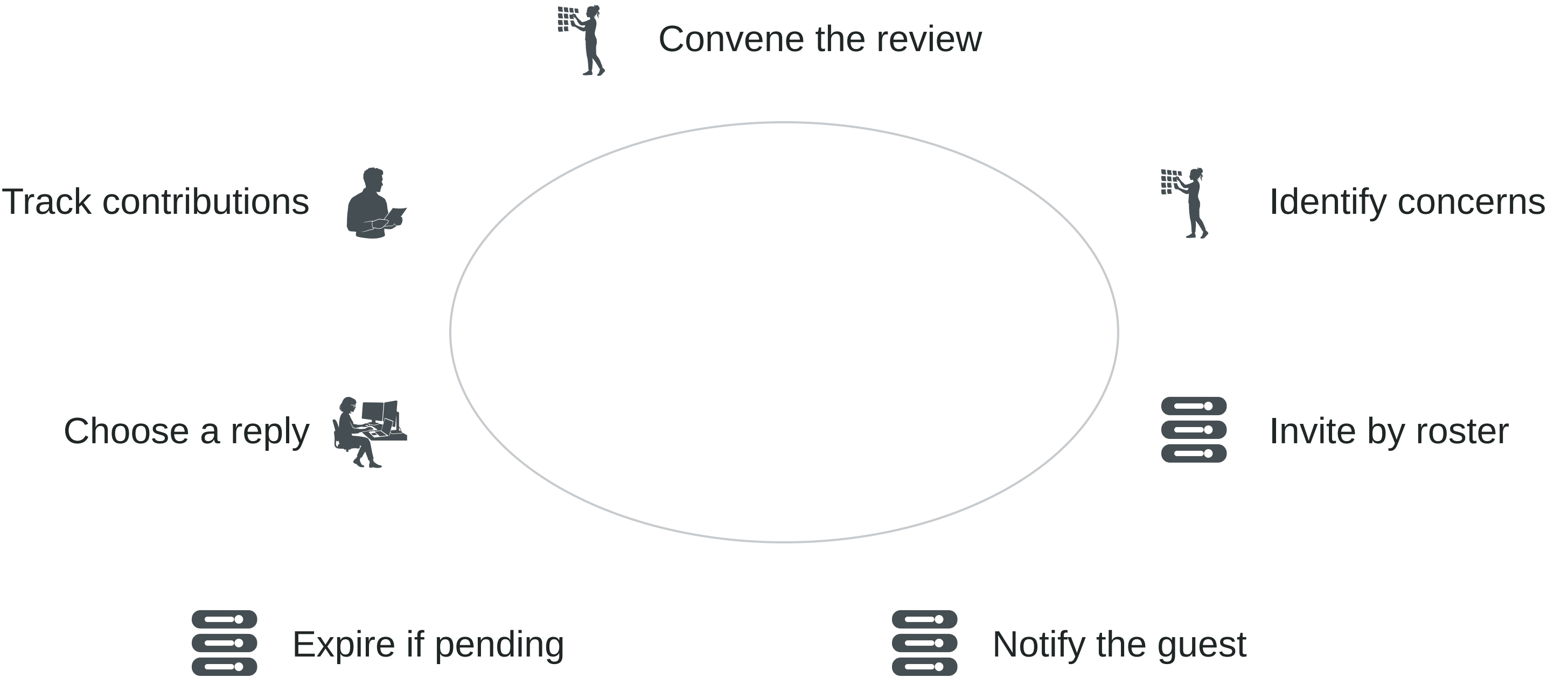
Austin's contribution
is absent from this record.

The next design needs a record of contributions.

EXERCISE · DELEGATION RELAY

How would you divide responsibilities
between people and software?

The work in this system



5 · The work we choose to delegate

People choose



Convene
the review



Choose
a reply

Interpretation + upkeep



Identify
concerns



Track
contributions

Rules carry it out



Invite
by roster



Notify the guest
Expire if pending

Maya and Sam do this now.

What record would preserve this work?
What actor could help interpret the updates?

PART 4

Adding agents

What would make Austin findable?



Austin

Contributed to refunds.
Holds no roster position.

What would Maya need to look up?

What kind of concept
could keep that information?

What would make Austin findable?



Austin

Contributed to refunds.
Holds no roster position.

What would Maya need to look up?

 A contribution record

concern	refunds
person	Austin
contribution	working on the fix

WorkTracking [Concern, Party]

Purpose	Keep a current record of who is working on each concern, and in what way. This prevents duplicated work and uncertainty about whom to ask.
Principle	Sam records Austin’s work on refunds and updates it when the fix is built. Maya finds Austin for the review. Once the work is done, Sam clears the entry.

State

- a Followed set of Concerns
- a set of Entries with
 - a concern Concern
 - a party Party
 - a role String

Actions

- follow**(concern)
- note**(concern, party, role)
return (entry)
- update**(entry, role)
- clear**(entry)

Sam records Austin's contribution

Earlier, while Austin was working on the fix



Sam

`note(refunds, Austin,
"working on it")`



 WorkTracking · entry w17

concern	refunds
party	Austin
role	working on it

role describes the contribution.

Already followed: refunds

The contribution is now available for another actor to read.

The expanded lookup finds Austin

when cover(the review, refunds)

 Rostering

concern	refunds
holder	Ada

 WorkTracking

concern	refunds
party	Austin



Invite each distinct person found in either record.

Ada is found

Austin is found

Keeping the work record current

Austin posts about refunds over the week:

“I’m looking into the failed refunds.”

“I think I’ve found the cause.”

“My patch is merged. Ready for release.”

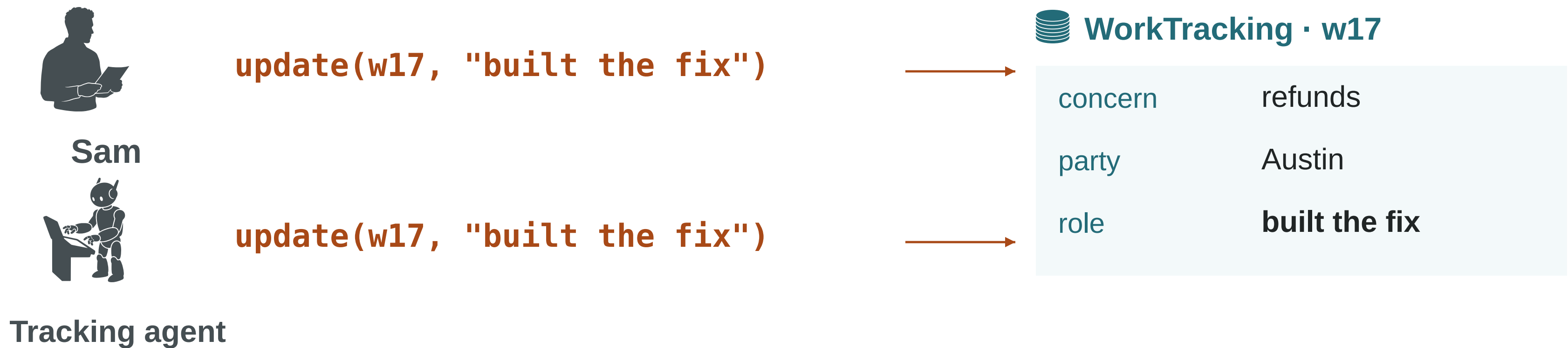


**Sam interprets
each update.**

Could an LLM help keep the description current?

An LLM can request the same update

Either Sam or an LLM (large language model) can request this update.



update changes the role. The concern and party stay the same.

The tracking agent's inputs

when `Discussing.post(refunds, text)` returns `post`

where · interpret the supplied facts

Task: keep Austin's work description current.

Austin's post in refunds

"My patch is merged.
Ready for release."

Which entry should
this update change?

Tool available

WorkTracking

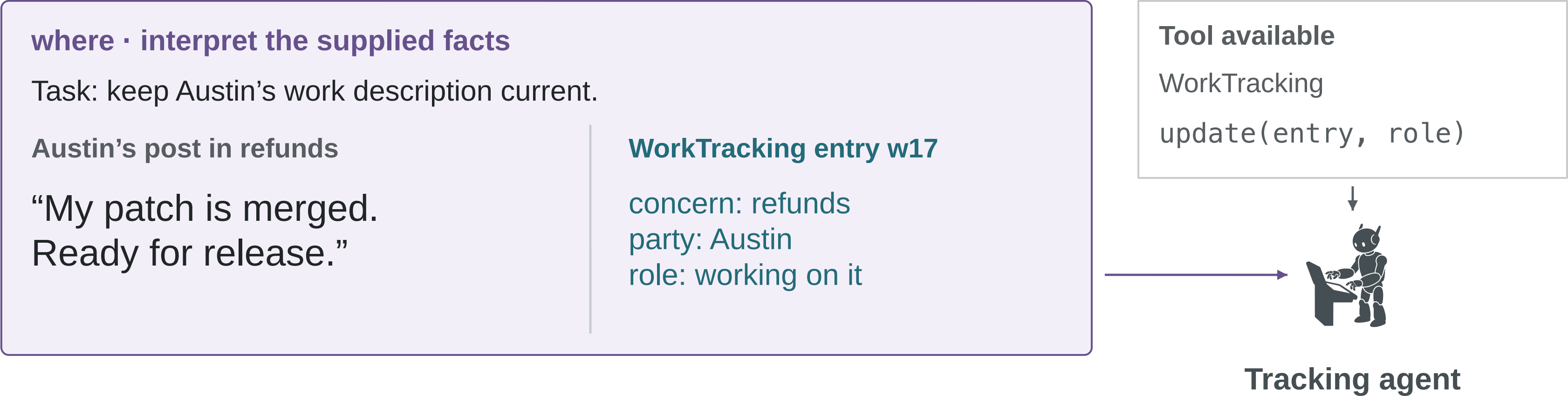
`update(entry, role)`



Tracking agent

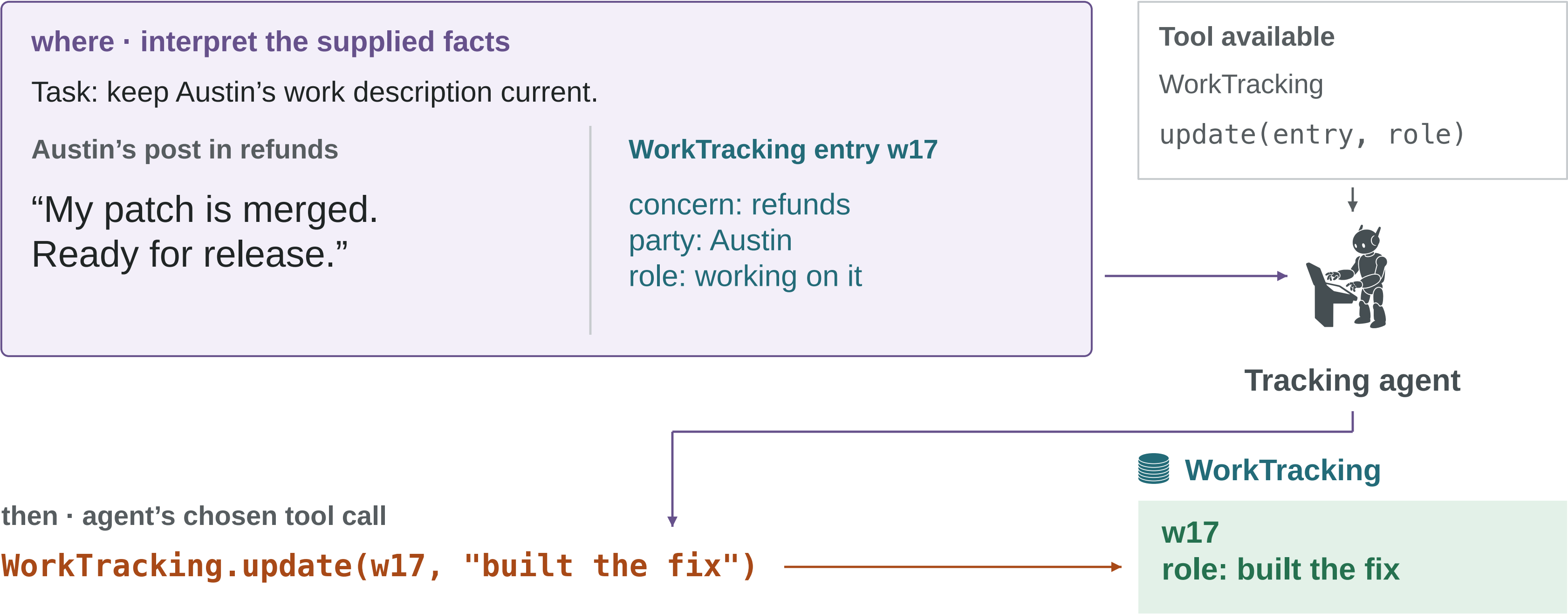
The entry is supplied with the post

when `Discussing.post(refunds, text)` returns `post`



The tracking agent's reaction

when `Discussing.post(refunds, text)` returns `post`



Interpreting a progress update

Suppose the agent interprets the same post differently.

Austin’s post

“My patch is merged.
Ready for release.”

Does the post support
what the record now claims?

 **WorkTracking · w17**

party	Austin
role	released

An allowed update can still be unsupported

Suppose the agent interprets the same post differently.

Austin's post

“My patch is merged.
Ready for release.”

 **WorkTracking · w17**

party	Austin
role	built the fix

The agent stored “released”. The post does not establish that.



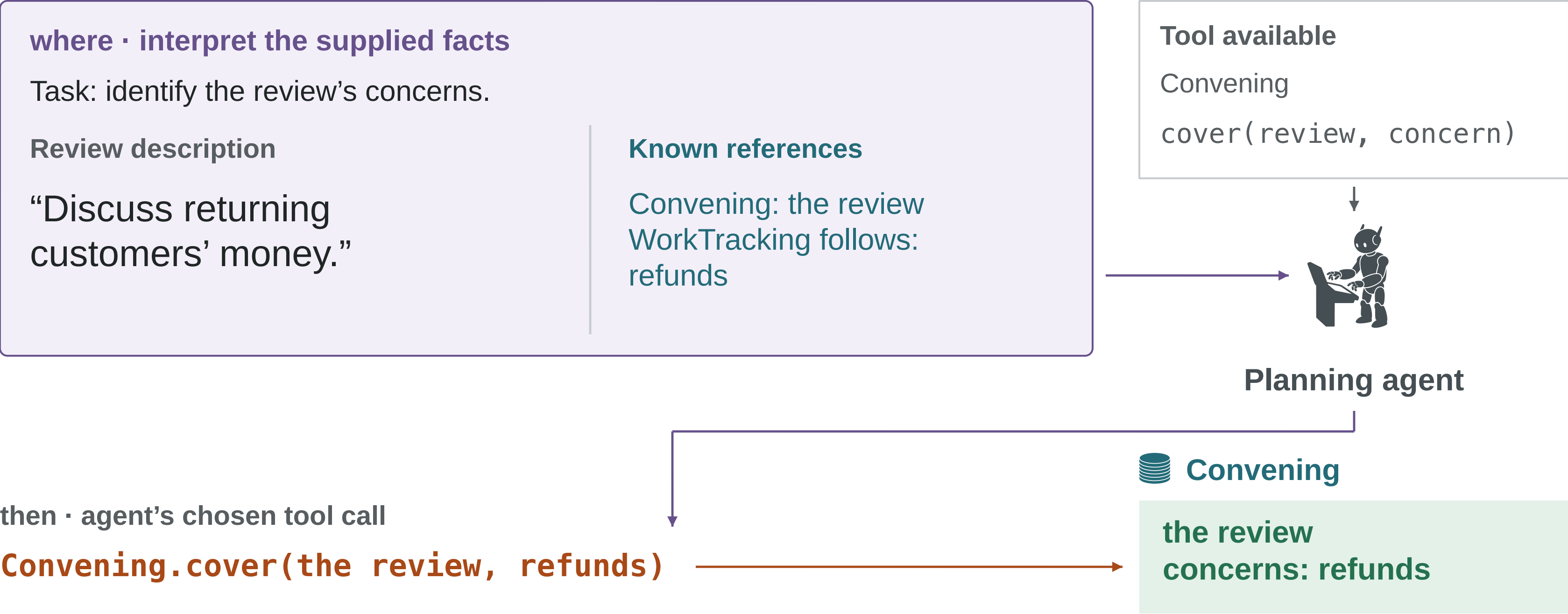
Sam corrects the record through the same action.

WorkTracking.update(w17, "built the fix")

An allowed action does not guarantee an accurate interpretation.

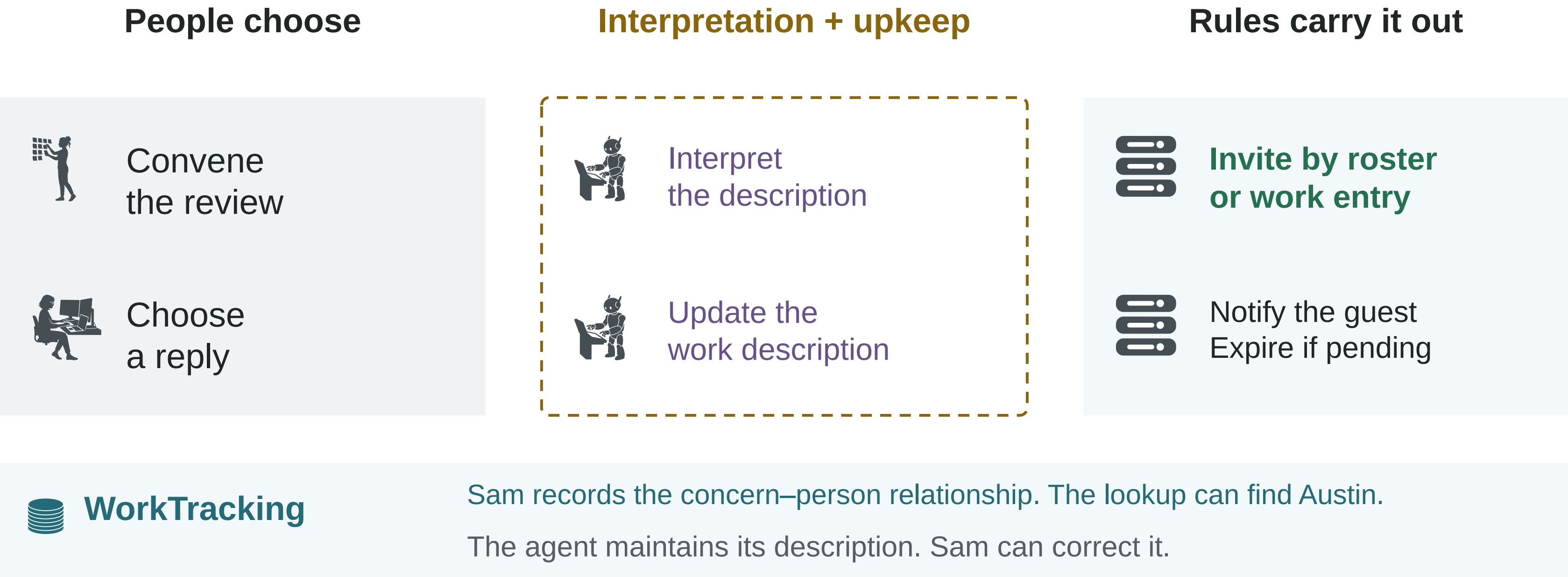
The planning agent identifies a concern

when `Convening.convene` returns the review



6 · People, machines, and agents coordinate

People keep their choices. Agents help with interpretation.



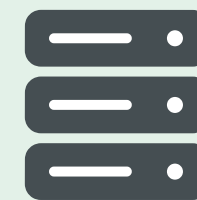
Austin can now contribute to the review

 WorkTracking

concern	refunds
party	Austin



`cover(the review, refunds)`



Austin

INVITATION RECEIVED

`invite
then notify`

Austin still chooses whether to accept.

The planning agent identifies refunds



Sam records entry w17

WorkTracking w17: concern = refunds, party = Austin
This relationship makes Austin findable.

“Discuss returning customers’ money.”
Followed concerns from WorkTracking: refunds



Maya

Convening

convene



Planning agent

Convening

cover



One chosen path. Orange: action handoffs. Blue dashed: record reads.

The lookup finds Austin's contribution



Sam records entry w17

WorkTracking w17: concern = refunds, party = Austin
This relationship makes Austin findable.

"Discuss returning customers' money."
Followed concerns from WorkTracking: refunds

For refunds:
Rostering → Ada
WorkTracking w17 → Austin
Review: Thursday



Maya

Convening

convene



Planning agent

Convening

cover



Lookup reaction

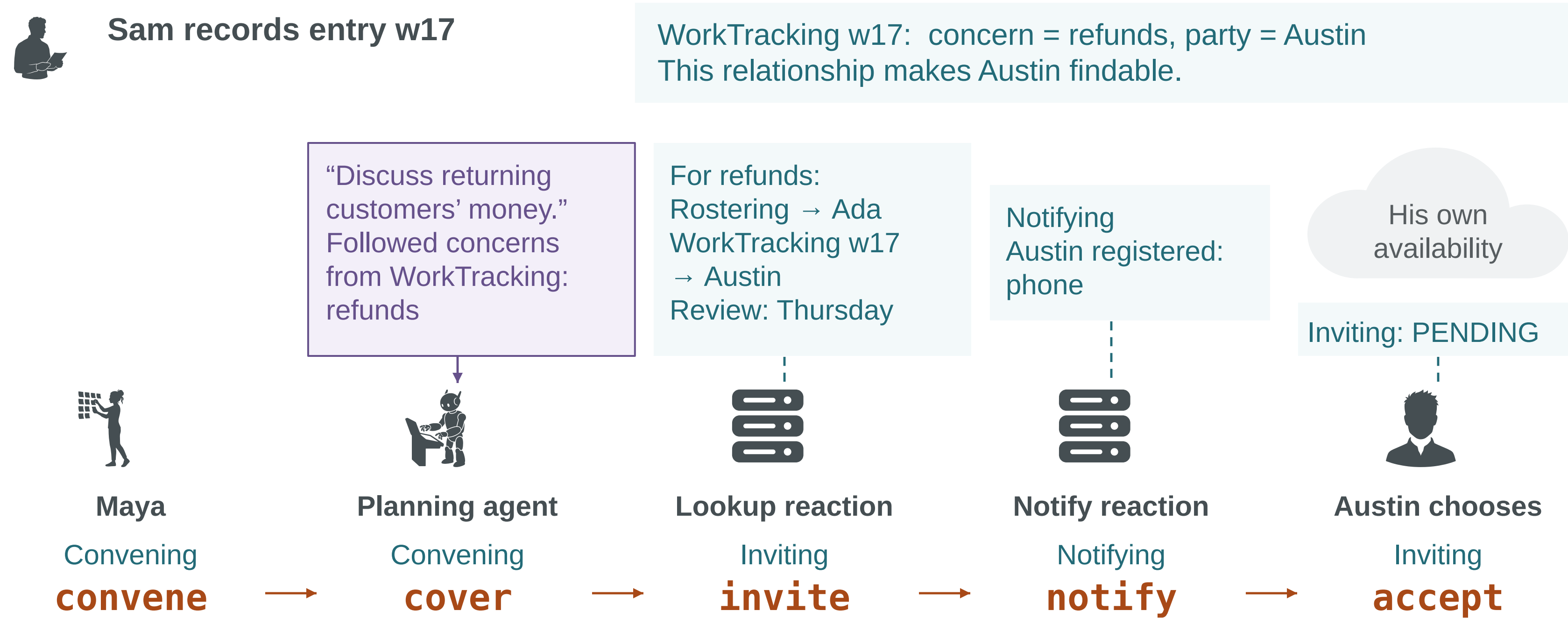
Inviting

invite



One chosen path. Orange: action handoffs. Blue dashed: record reads.

One invitation through the system



One chosen path. Orange: action handoffs. Blue dashed: record reads.

Lessons 1–3 · What did we learn?

1 Concepts

PENDING → ACCEPTED

2 Actions

```
invite(review, Ada,  
        Wednesday)
```

3 Reactions



Lessons 1–3 · State and coordination

1 Concepts

PENDING → ACCEPTED

Keep and govern state

2 Actions

`invite(review, Ada,
Wednesday)`

Carry shared meaning

3 Reactions



Connect through actions

Lessons 4–6 · What did we learn?

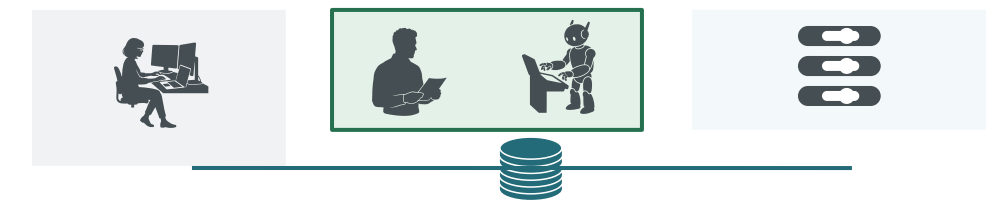
4 Reads



5 Delegation



6 Agent extension



Lessons 4–6 · Context and delegation

4 Reads



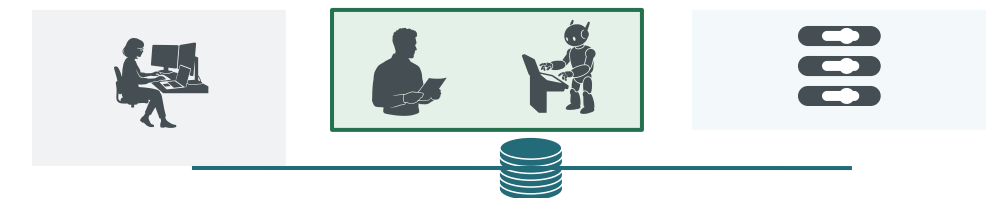
Supply local context

5 Delegation



Assign work and its upkeep

6 Agent extension



Add records and reactions

Try it on a system you know

Choose a system you know.

For example: someone requests access.

What event starts this?

One event someone can notice

Who acts?

Person, LLM, or machine

What do they read?

A record, a message, other knowledge

What do they ask for?

One meaningful action

Which needed fact is missing? Who will supply it?